

## CORRESPONDENCE OPEN



## Orexin receptor antagonists for major depressive disorder: perspectives from a systematic review and meta-analysis

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Dear Editor

A recent report by Dr. Hashimoto in *Molecular Psychiatry* highlighted that, because the orexin system in mesolimbic pathways modulates reward and motivation, it is important to monitor mood changes in individuals with insomnia treated with orexin receptor antagonists (QXR-ANTs) [1]. Orexin receptors comprise two subtypes: OX1R and OX2R.

Recent double-blind, randomized, placebo-controlled trials (DBRPCTs) showed that seltorexant, a selective OX2R antagonist, alleviated overall symptoms in individuals with major depressive disorder (MDD) [2]. Nevertheless, other QXR-ANTs (filorexant and tebideutorexant) did not improve symptoms in individuals with MDD in DBRPCTs, leaving their therapeutic value uncertain [3, 4]. Therefore, we conducted a systematic review and meta-analysis to evaluate the efficacy, acceptability, tolerability, and safety of QXR-ANTs in individuals with MDD.

This study followed Preferred Reporting Items for Systematic reviews and Meta-Analyses guidelines (Table S1) [5]. The PICO framework was defined as follows: Patients, adults with MDD; Intervention, pooled QXR-ANTs; Comparison, placebo; Outcome, improvement in overall symptom scores (17 items in the Hamilton Depression Rating Scale [HDRS17] [6] and Montgomery-Åsberg Depression Rating Scale [MADRS] [7], primary), response rate ( $\geq 50\%$  reduction in HDRS17 or MADRS total score from baseline), remission rate (HDRS17 total score  $\leq 7$  or MADRS  $\leq 12$ ), all-cause discontinuation, adverse event-related discontinuation, and incidence of individual adverse events. We searched PubMed, the Cochrane Library, and Embase databases for studies published before October 26, 2025. The detailed strategy is provided in Figure S1. A meta-analysis was performed for outcomes reported in at least two DBRPCTs. Because two trials included multiple seltorexant dose arms [2, 8], these were pooled to avoid unit-of-analysis error [9]. Data were analyzed using intention-to-treat or modified intention-to-treat principles; data based on completion analyses were excluded.

A random-effects [10], pairwise meta-analysis was conducted in Review Manager software [9]. Standardized mean differences (SMD) and risk ratios (RR) were calculated for continuous and dichotomous variables, respectively, with 95% confidence intervals (95% CIs). Heterogeneity was assessed using  $I^2$ , where  $I^2 \geq 50\%$  indicated substantial heterogeneity [9]. MADRS and HDRS17 were used in a filorexant study [3], while one seltorexant study used only MADRS [8]. Two other seltorexant studies [2, 11] and a tebideutorexant study [4] used only HDRS17. HDRS17 includes three sleep-related items, whereas MADRS includes only one, and some QXR-ANTs are approved for insomnia [12]. Thus, the choice of rating scale may have influenced our results. Therefore, primary outcome utilized HDRS17 data in preference to MADRS data ("overall symptoms [HDRS17]"), our meta-analysis

also included outcome of "overall symptoms (MADRS)" that used MADRS data in preference to HDRS17 data as a secondary efficacy outcome. Publication bias was assessed using Egger's test (Comprehensive Meta-Analysis, version 3 [Biostat Inc., Englewood, NJ, USA]). Statistical significance was defined as two-sided  $p < 0.05$ .

Figure S1 illustrates study selection. Six studies met the inclusion criteria, but because results from NCT02669030 (suvorexant) [13] were not reported, the meta-analysis included one filorexant DBRPCT [3], three seltorexant DBRPCTs [2, 8, 11] and one tebideutorexant DBRPCT [4]. Study characteristics and pharmacological profiles of each QXR-ANT are presented in Table S2 and in Text S1, respectively. Four studies were evaluated as "low risk" for overall risk of bias (Table S3) [9].

Pooled QXR-ANTs demonstrated a significantly greater reduction in overall symptom scores than placebo. For "overall symptoms (HDRS17)" and "overall symptoms (MADRS)", the SMDs (95% CIs) were  $-0.16$  ( $-0.30$  to  $-0.01$ ),  $p = 0.04$ ,  $I^2 = 0\%$  (Fig. 1), and  $-0.16$  ( $-0.31$  to  $-0.01$ ),  $p = 0.03$ ,  $I^2 = 0\%$  (Figure S2), respectively. Egger's test showed no significant publication bias ( $p = 0.17$ ). Pooled QXR-ANTs yielded a higher remission rate than placebo (RR [95% CI] =  $1.52$  [ $1.05$ – $2.20$ ],  $p = 0.03$ ,  $I^2 = 0\%$ ), but showed no significant differences for other outcomes (Table S4).

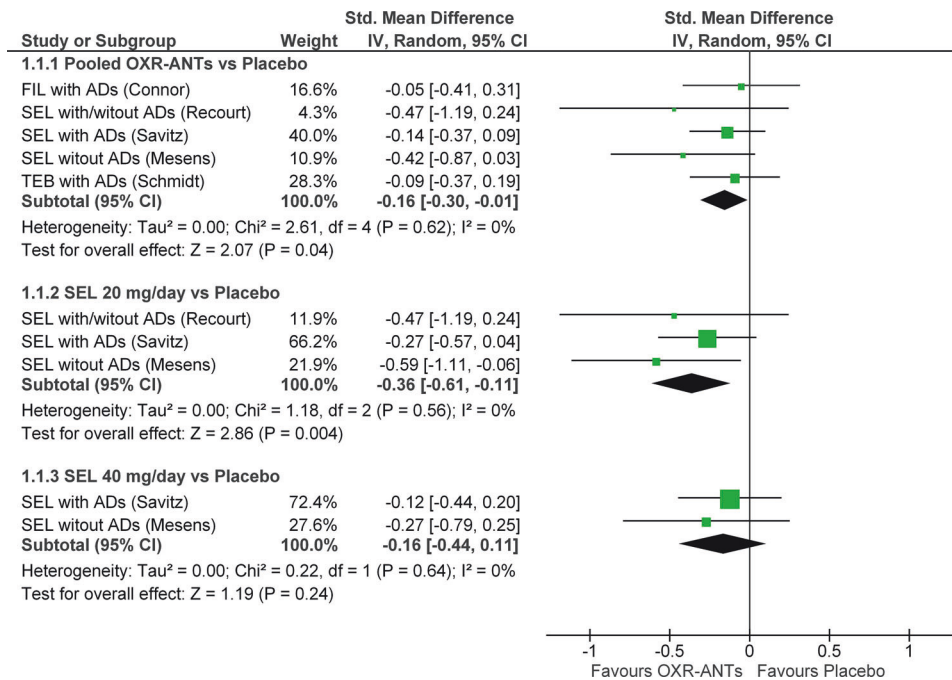
Subgroup analysis by seltorexant dose showed that 20 mg/day significantly improved overall symptom scores compared with placebo (SMD [95%CI] =  $-0.36$  [ $-0.61$ ,  $-0.11$ ],  $p = 0.004$ ,  $I^2 = 0\%$ ), whereas 40 mg/day did not (Fig. 1). Seltorexant 20 mg/day was marginally associated with higher remission rates (RR [95% CI] =  $1.61$  [ $1.00$ – $2.58$ ],  $p = 0.05$ ,  $I^2 = 0\%$ ), but showed no significant differences for other outcomes (Table S5). Seltorexant 40 mg/day was not superior to placebo for any efficacy outcome, but was marginally associated with higher adverse event-related discontinuation (RR [95% CI] =  $5.27$  [ $0.99$ – $27.91$ ],  $p = 0.05$ ,  $I^2 = \text{not applicable}$ ), and was associated with a higher incidence of nightmares (RR [95% CI] =  $8.77$  [ $1.03$ – $74.58$ ],  $p = 0.05$ ,  $I^2 = 0\%$ , Table S5).

Three sensitivity analyses were conducted for the primary outcome (Table S6). The fixed-effect model yielded results consistent with the primary analysis. However, when excluding the smallest trial [11] and using the MADRS alone, the superiority of QXR-ANTs was not observed, likely reflecting type II error due to limited sample size.

Approximately 55% of people with MDD meet criteria for inadequate response to  $\geq 2$  antidepressants despite adequate treatment [14]. Although second-generation antipsychotics are used in such cases, their long-term safety remains uncertain [15]. Therefore, it is imperative to develop novel treatment options. Our meta-analysis indicates that pooled QXR-ANTs, particularly seltorexant 20 mg/day, provide clinical benefit for MDD and show favorable acceptability, tolerability, and safety. Notably, our meta-analysis shows that QXR-ANTs did not increase the risk of somnolence, dizziness, or abnormal dreams. Nonetheless, clinicians should remain vigilant and carefully manage these potential

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**Fig. 1 Forest Plots.** 95% CI 95% confidence interval, ADs antidepressants, FIL filorexant; IV inverse variance, OXR-ANTs orexin receptor antagonists, SEL seltorexant, Std standard, TEB tebideutorexant.

adverse events when administering QXR-ANTs. In addition, several limitations of our meta-analysis warrant consideration (Text S2).

Our meta-analysis showed that seltorexant 20 mg/day, but not 40 mg/day, was effective in individuals with MDD, possibly reflecting differences in tolerability and safety (Table S5). Neither filorexant nor tebideutorexant showed antidepressant efficacy [3, 4]. Higher placebo responses in those trials may partly explain this pattern, and differences in receptor antagonistic profiles may also contribute. Seltorexant, an OX2R antagonist, is expected to modulate the hypothalamic–pituitary–adrenal axis, which is considered one of the key pathophysiological mechanisms underlying MDD [2, 8, 11]. However, it remains unclear how selective antagonism at OX1R or OX2R, or dual antagonism at both receptors, may differentially affect efficacy and safety in MDD, as shown in Text S1.

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## DATA AVAILABILITY

Data used for the current study were reported in articles as cited in this paper.

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## AUTHOR CONTRIBUTIONS

TK had full access to all the data in the study and takes responsibility for the integrity of the data along with the accuracy of the data analysis. TK was involved in the study concept and design. The literature search, data transfer accuracy, and acquisition and calculations of data were conducted by at least two of the authors (TK, KS, MH, SH, and YN). TK performed the statistical analysis. All the authors performed interpretation of the data, and wrote the manuscript. NI supervised the review.

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## COMPETING INTERESTS

All authors have no conflicts of interest to declare concerning this study. They also declare any potential competing interests that have arisen in the last 3 years. Dr. Kishi has received speaker's honoraria from Daiichi Sankyo, Shionogi, Eisai, Janssen, Meiji, Otsuka, Sumitomo, Takeda, Mitsubishi-Tanabe, Kyowa, Boehringer Ingelheim, and Viartis and research grants from Eisai, JSPS KAKENHI (23K06998 and 25K10874), the Japan Agency for Medical Research and Development (JP22dk0307107, JP22wm0525024, 23dk0307117h0001, and 24dk0307129h0001, 256f0137005j0001, 266f0137005j0002, and 276f0137005j0003), and the Japanese Ministry of Health, Labour and Welfare (21GC1018). Dr. Sakuma has received speaker's honoraria from Janssen, Kyowa, Meiji, Otsuka, Sumitomo, and Takeda and research grants from JSPS KAKENHI (19K17099 and 23K06998) and Japan Agency for Medical Research and Development (JP22dk0307107 and JP23dk0307122). Dr. Hatano received the speaker's honoraria from Meiji and Sumitomo, and has received Grant-in-Aid for Early-Career Scientists (23K14827). Dr. Hamanaka has received speaker's honoraria from Meiji, Otsuka, and Sumitomo. Dr. Nishii has received speaker's honoraria from Meiji, Otsuka, and Sumitomo. Dr. Iwata has received speaker's honoraria from Eisai, Janssen, Meiji, Otsuka, Sumitomo, Takeda, Mitsubishi-Tanabe, and Viartis and research grants from Daiichi Sankyo, Eisai, Meiji, Otsuka, Sumitomo, Takeda, Tanabe-Mitsubishi, Grant-in-Aid for Scientific Research (B)(22H03003), and Japan Agency for Medical Research and Development (JP22wm0425008).

## ETHICS APPROVAL AND CONSENT TO PARTICIPATE

The current study includes a systematic review and meta-analysis based on secondary data from previously published, publicly available data. No new human

or animal participants were recruited or studied directly by the authors. This study was performed in accordance with the relevant guidelines and regulations. All studies included in this systematic review and meta-analysis had received ethical approval from their respective institutional review boards, and informed consent was obtained from all participants in those original studies. Therefore, no additional ethical approval or informed consent was required for this study.

## ADDITIONAL INFORMATION

**Supplementary information** The online version contains supplementary material available at <https://doi.org/10.1038/s41380-026-03599-z>.

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